

Technology Skill Development Lab Java

Assignment No.3

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Class : SY

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Problem Statement : Create a base class Person with attributes name and age. Include a method display_info() to print basic information about the person.

a.Single Inheritance: Derive a class Student from the Person class with additional attributes student_id and major. Include a method enroll_course() to simulate a student enrolling in a course.

b. Multilevel Inheritance: Extend the Student class to create a new class GraduateStudent with an additional attribute research_topic. Implement a method conduct_research() to simulate a graduate student conducting research.

c. Hierarchical Inheritance: Derive a class Professor from the Person class with attributes like employee_id, department, and teaching_subject. Implement a method conduct_lecture() to simulate a professor conducting a lecture.

Program:

```
class Person {

  String name = "Saurabh";

  int age = 19;

  void display_info() {

     System.out.println("Name of the person: " + name);

     System.out.println("Age of the person: " + age);

  }

}

// Single Inheritance

class student extends Person {

  int student_id = 1234;

  String major = "AI & DS";
```

```
    void enroll_course() {
        System.out.println(name + " has enrolled in " + major + ".");
    }
}
// Multilevel Inheritance
class Graduate_student extends student {
    String research_topic = "AI ERA";
    void conduct_research() {
        System.out.println("Graduate student conducting research on: " + research_topic);
    }
}
// Hierarchical Inheritance
class Professor extends Person {
    int employee_id = 5678;
    String department = "Computer Science";
    String teaching_subject = "Java";
    void conduct_lecture() {
        System.out.println("Employee ID: " + employee_id);
        System.out.println("Department: " + department);
        System.out.println("Teaching Subject: " + teaching_subject);
        System.out.println(name + " is conducting a lecture.");
    }
}
```

```
public class inherit {
   public static void main(String[] args) {
      // Student
      student s1 = new student();
      System.out.println("STUDENT");
      s1.display_info();
      System.out.println("Student ID: " + s1.student_id);
      System.out.println("Major: " + s1.major);
      s1.enroll_course();
      System.out.println();
      // Graduate Student
      Graduate_student s2 = new Graduate_student();
      System.out.println("GRADUATE STUDENT");
      s2.display_info();
      System.out.println("Student ID: " + s2.student_id);
      System.out.println("Major: " + s2.major);
      s2.enroll_course();
      System.out.println("Research Topic: " + s2.research_topic);
      s2.conduct_research();
      System.out.println();
      // Professor
      Professor p1 = new Professor();
      System.out.println("PROFESSOR");
```

```
        p1.display_info();
        System.out.println("Employee ID: " + p1.employee_id);
        System.out.println("Department: " + p1.department);
        System.out.println("Teaching Subject: " + p1.teaching_subject);
        p1.conduct_lecture();
    }
}
```

Output :

```
PS C:\Users\VUstudent\saurabhhhh.java> cd "c:\Users\VUstudent\saurabhhhh.java\"
STUDENT
Name of the person: Saurabh
Age of the person: 19
Student ID: 1234
Major: AI & DS
Saurabh has enrolled in AI & DS.

GRADUATE STUDENT
Name of the person: Saurabh
Age of the person: 19
Student ID: 1234
Major: AI & DS
Saurabh has enrolled in AI & DS.
Research Topic: AI ERA
Graduate student conducting research on: AI ERA

PROFESSOR
Name of the person: Saurabh
Age of the person: 19
Employee ID: 5678
Department: Computer Science
Teaching Subject: Java
Employee ID: 5678
Department: Computer Science
Teaching Subject: Java
Saurabh is conducting a lecture.
PS C:\Users\VUstudent\saurabhhhh.java>
```

Problem Statement : Design a multilevel inheritance system to handle dataset processing. Implement a Data Cleaning Algorithm to remove invalid records such as negative marks or empty names.

Program:

```
class Dataset {
  String[] names = {"Saurabh", "", "Rahul", "Ruturaj", "Amit"};
  int[] marks = {85, 90, -10, 75, -5};
  void display_data() {
    System.out.println("Dataset:");
    for (int i = 0; i < names.length; i++) {
      System.out.println(names[i] + " : " + marks[i]);
    }
  }
}
// Level 2
class DataProcessor extends Dataset {
  void process_data() {
    System.out.println("Processing dataset...");
  }
}
// Level 3
class DataCleaner extends DataProcessor {
  void clean_data() {
    System.out.println("\nValid Records:");
```

```
        for (int i = 0; i < names.length; i++) {
            // Remove empty names and negative marks
            if (!names[i].isEmpty() && marks[i] >= 0) {
                System.out.println(names[i] + " : " + marks[i]);
            }
        }
    }
}
public class data {
    public static void main(String[] args) {
        DataCleaner d = new DataCleaner();
        d.display_data();
        d.process_data();
        d.clean_data();
    }
}
```

Output :

```
PS C:\Users\VUstudent\saurabhhhh.java> cd "c:\Users\VUstudent\saurab
Dataset:
Saurabh : 85
 : 90
Rahul : -10
Ruturaj : 75
Amit : -5
Processing dataset...

Valid Records:
Saurabh : 85
Ruturaj : 75
PS C:\Users\VUstudent\saurabhhhh.java>
```

Conclusion: Thus, the concepts of different types of inheritance and dataset processing were successfully implemented in Java. The program demonstrates code reusability through inheritance and performs basic data cleaning by removing invalid records.